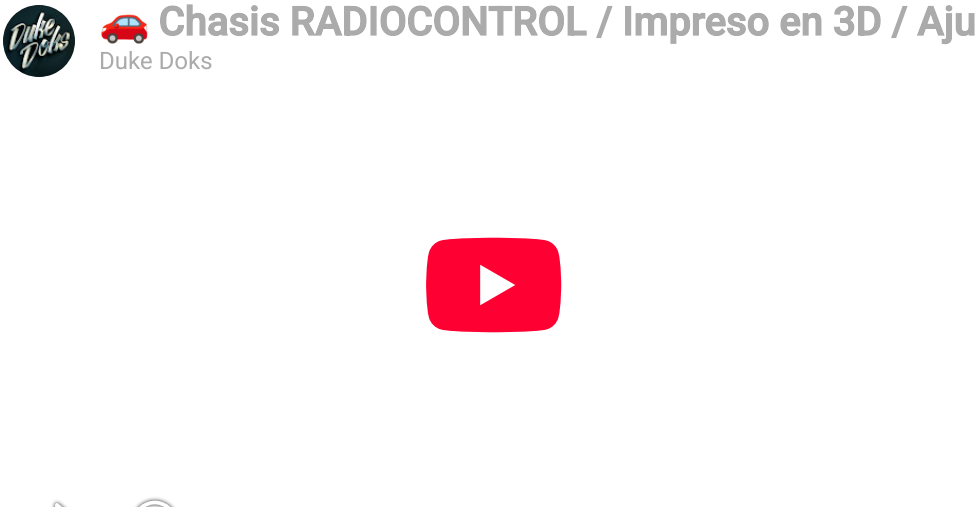
DKS-Basic Chassis

RESOURCES

Construction Guide

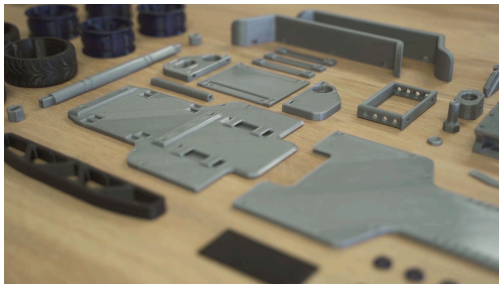
The DKS-Basic is a personal project where I designed a 1:10 scale universal radio-controlled chassis. The wheelbase is adjustable and it works with both commercially available bodies and those I design myself, such as the Dodge Challenger, [https://dukedoks.com/portfolio/guia-dodge-challenger/] which you'll also find information about on this website. On this page, you'll find all the information and everything you need to build and replicate the project yourself. You'll find links to the STL files, electronic components, filament, instructions, and a video of the process. Don't forget to subscribe to my YouTube channel—it's free! And if you like this or any of my other projects, please feel free to share it. Thank you, and I hope you enjoy it.

SUBSCRIBE



STL files

I usually share all my STL design files on sites like Cults, Printables, and Makerworld. In this case, you can find it in all three repositories completely free of charge. To download and start printing this design, simply click one of the buttons below.



DOWNLOAD MAKERWORLD FILES

DOWNLOAD PRINTABLE FILES

DOWNLOAD CULTS FILES

Recommended print settings

These are the settings I used, but some parameters depend on the type of material you're using. I used PLA filament from Fillamentum and Sakata 3D, and TPU from Smartmaterials for the tires. You can start with these settings and adjust them as needed depending on the material you're using. You'll find all the filaments I used further down this page.

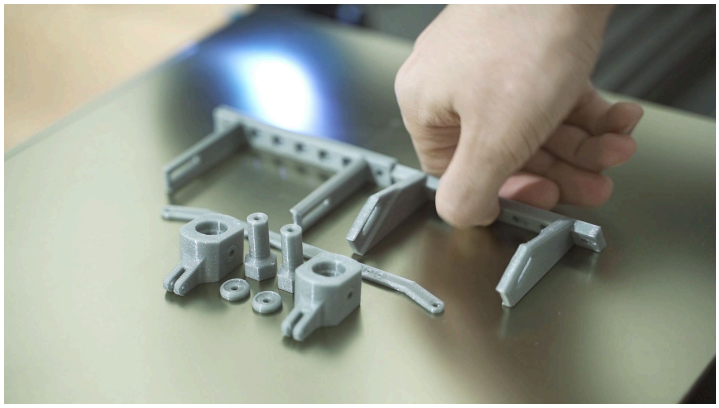
All files are already positioned for printing, meaning they will print in their optimal orientation when imported into your slicer. No parts require supports except for the "Basic Wheel 01 Wheel" rims. While there is also a version without supports, this version includes the support already created in the design, so it won't need to be activated in the slicer. The file is "Basic Wheel 01 Wheel no support".

Chassis and wheels in PLA

Flexible TPU tires

- Layer height: 0.2 mm
- Wall thickness: 1.2 mm
- Top and bottom layers: 5
- Filling density: 6%
- Fill pattern: Gyroid
- Nozzle temperature: 220º
- Bed temperature: 50º
- Retraction: On
- Flow: 100%
- Print speed: 55

- Layer height: 0.2 mm
- Wall thickness: 1.2 mm
- Upper and lower layers: 4
- Filling density: 0%
- Fill pattern: Gyroid
- Nozzle temperature: 230º
- Bed temperature: 50º
- Retraction: Off
- Flow: 110%
- Print speed: 25



Recommended print settings

These are the settings I used, but some parameters depend on the type of material you're using. I used PLA filament from Fillamentum and Sakata 3D, as well as recycled TPU from Lowpoly for the tires. You can start with these settings and adjust them as needed depending on the material you're using. You'll find all the filaments I used further down this page.

All files are already positioned for printing, meaning they will print in their optimal orientation when imported into your slicer. Generally, no parts require supports except for the "Basic Wheel 01 Wheel" rims. While you'll also find a version without supports, in this version, the support is already created in the design, so it won't need to be activated in the slicer. The file is "Basic Wheel 01 Wheel no support".

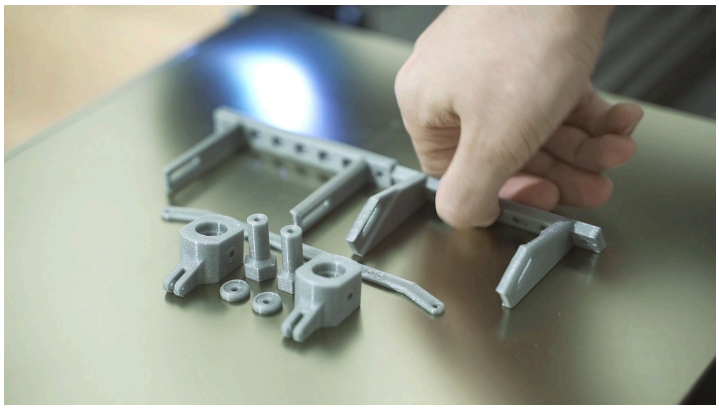
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Flexible TPU tires

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- Filling density: 0%
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- Nozzle temperature: 230º
- Bed temperature: 50º
- Retraction: Off
- Flow: 110%
- Print speed: 25



Components and tools

All the components have buttons below that take you to another website to buy them, such as Amazon, Banggood, AliExpress, 3D Printers, etc. These are affiliate links,

which means that if you buy through them, you'll pay exactly the same price and I'll receive a small commission. This way, you're supporting this project and future ones.

For this project, we'll need the components listed below, although some are optional or alternatives to others. Below is a list of the essential components, but I'll also explain the different possibilities.

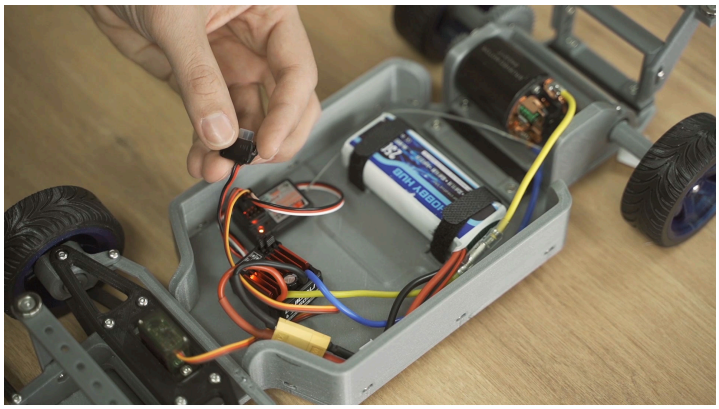
Motor and speed controller: I used a brushed motor and speed controller; they're cheaper and I prefer them for this project. But I've also included an alternative brushless kit below. These are more expensive, but they're more efficient, more powerful, and suffer less wear.

Servo: In the video, I show how to install a small servo, but with use, I've realized it's a bit underpowered, especially when adding a heavy body. If you use these servos, at least make sure they have metal gears, or they'll break very quickly. I currently have two steering systems designed for the chassis: the one I show in the video and a second one I designed later that uses a larger servo. I recommend printing the latter. Depending on which one you choose, you'll need to buy a different servo.

Transmitter: I'm also including two transmitter models. The best one for this project is the pistol-grip type, designed for cars and works great. It also has memory for 10 receivers. The other transmitter is recommended if you want to create other types of radio-controlled projects in the future; it has more channels and could also be used for drones or airplanes, for example. But both are suitable for this project.

Wheels: Both the rims and tires are printable, but the TPU I use for the tires doesn't offer much grip when driving the chassis. So, if you want to get the most out of the driving experience, I've also included some wheels. Mine are designed based on the measurements of commercially available wheels, so you can use the printed rims with the commercial tire, or use the complete commercial wheel on my chassis.

- X2 – M3 5mm Screw
- X39 – M3 8mm Screw
- X20 – M3 12mm Screw
- X2 – M3 16mm Screw
- X8 – M3 8mm countersunk head screw
- X38 – M3 Nuts
- X6 – M3 Washers
- X8 – Magnets 12x3x3mm
- X2 – Bearing 12x18x4mm



- X4 – Bearing 8x12x3.5mm
- Engine
- Variable speed drive
- Servo
- Battery
- Charger
- Radio Station



Small Servo



Servo MG995 180º



M3 screws



Brushed Variator

BUY ON AMAZON

BUY ON AMAZON

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BUY ON AMAZON

SHOP AT BANGGOOD

BUY ON AMAZON

BUY ON ALIEXPRESS

SHOP AT BANGGOOD

BUY ON ALIEXPRESS

SHOP AT BANGGOOD

BUY ON ALIEXPRESS

BUY ON ALIEXPRESS



Brushed motor



Battery



Lipo Charger



M3 Nuts

BUY ON AMAZON	BUY ON AMAZON	BUY ON AMAZON	BUY ON AMAZON
SHOP AT BANGGOOD	SHOP AT BANGGOOD	SHOP AT BANGGOOD	SHOP AT BANGGOOD
BUY ON ALIEXPRESS	BUY ON ALIEXPRESS	BUY ON ALIEXPRESS	BUY ON ALIEXPRESS



Bearings 12x18x4mm



Bearings 8x12x3.5mm



3-channel broadcaster



Station with 10 channels

BUY ON ALIEXPRESS

BUY ON ALIEXPRESS

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BUY ON AMAZON

BUY ON AMAZON

BUY ON AMAZON

SHOP AT BANGGOOD

SHOP AT BANGGOOD

SHOP AT BANGGOOD

BUY ON ALIEXPRESS

BUY ON ALIEXPRESS



Wheels



XT60 connector



T-connector to XT60 converter



Speed controller and brushless motor

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BUY ON AMAZON

BUY ON AMAZON

BUY ON AMAZON

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SHOP AT BANGGOOD

SHOP AT BANGGOOD

SHOP AT BANGGOOD

BUY ON ALIEXPRESS

BUY ON ALIEXPRESS

BUY ON ALIEXPRESS

BUY ON ALIEXPRESS



Magnets



Bodywork

BUY ON AMAZON

BUY ON AMAZON

BUY ON ALIEXPRESS

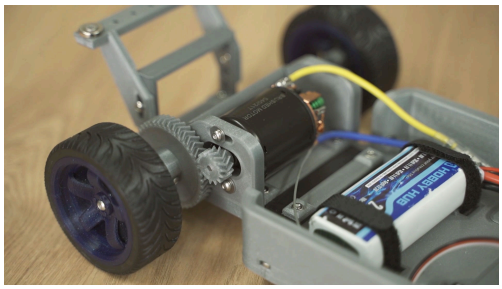
SHOP AT BANGGOOD

BUY ON ALIEXPRESS

Filament

Different filaments are used for this project; here are the ones I used, but you can use whichever you prefer. Just keep in mind that you should use a flexible filament for the tires—we want them to be soft. Regarding

flexible filaments for the tires, I'll give you two options. In the video, I use the one from Smartmaterials, but I recently discovered the one from Lowimpact, which is made from recycled tires and has proven to have better grip.

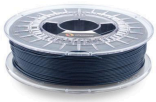


Another tip I could give you is to print the parts that experience the most stress or wear in a different material. I've used PLA 850 and it's worked well for me. But for example, it would be a good idea to print the gears in ABS, ASA, or some other technical material.



PLA Sakata 850

BUY



PLA Fillamentum

BUY



Flexible Smartfil

BUY



Flexible Lowimpact

BUY

BUY ALTERNATIVE

BUY ALTERNATIVE

BUY ALTERNATIVE

BUY ALTERNATIVE

Mounting

Here's a step-by-step guide to the assembly. For each step, I'll tell you the number of screws, the components, and the tools you'll need. Get your tools ready and let's begin.



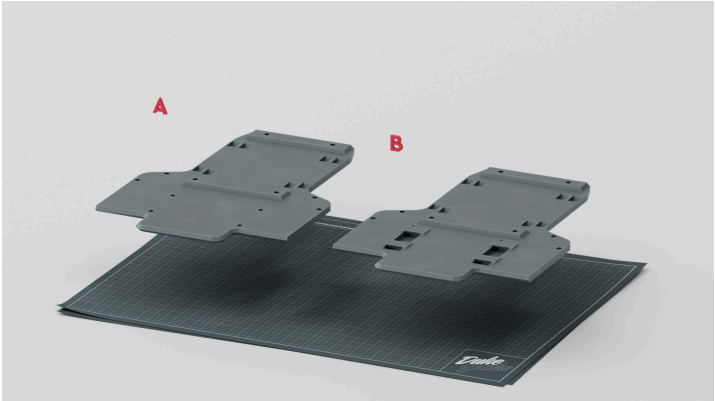
Step 1

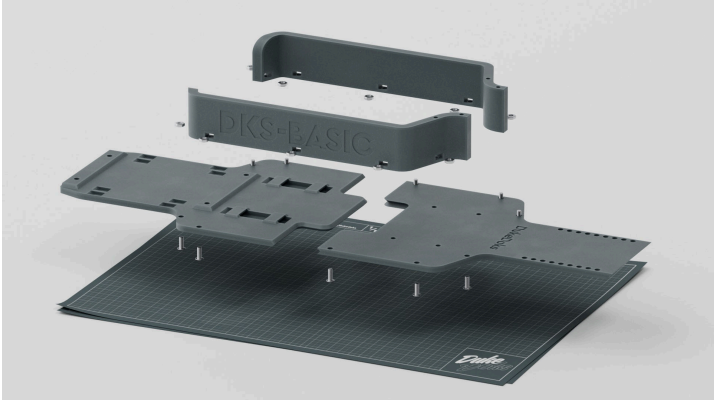
Components: 10 nuts and 10 M3 8mm screws.

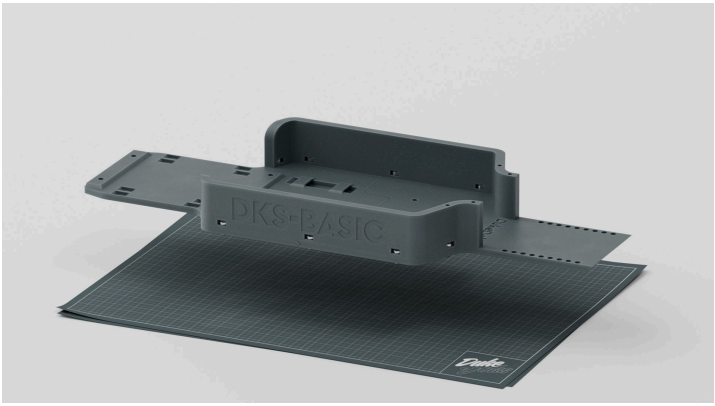
Printed parts: “Chassis-1”, “Chassis-2_A” or “Chassis-2_B”, “Body_R” and “Body_L”.

Main Body. We fit “Chassis-1” and “Chassis-2” together and then attach “Body_R” and “Body_L” to the previous assembly with 10 nuts and 10 M3 8mm screws.

Among the chassis components, there's one part that comes in two versions. The only difference is that version "B" has slots for inserting Velcro straps to secure the battery, while version "A" has four 3mm holes in case you need to design an additional part to attach your components. These holes allow you to add screws and fix your part in place.







Step 2

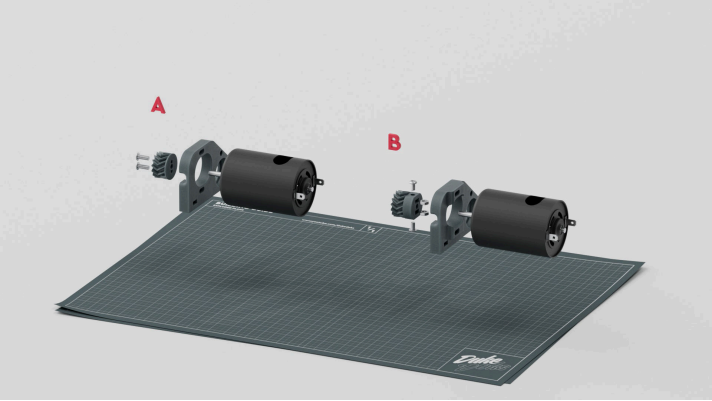
Components: 10 M3 screws (8mm), 8 M3 nuts, and 2 bearings (12x18x4mm)

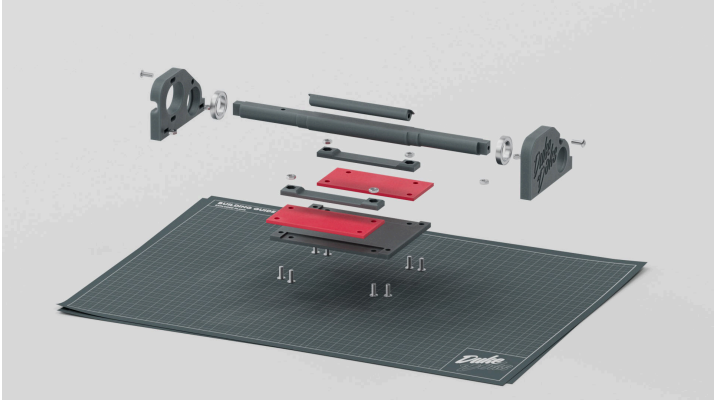
Printed parts: “Rear-Frame_L”, “Rear-Frame_R_A” or “Rear-Frame_R_B”, “Rear-Frame-Lower”, “Rear-Axis”, “Clamp” x2, “Rear-Flex-1”, “Rear-Flex-2” and “Rear-Frame-Backing”

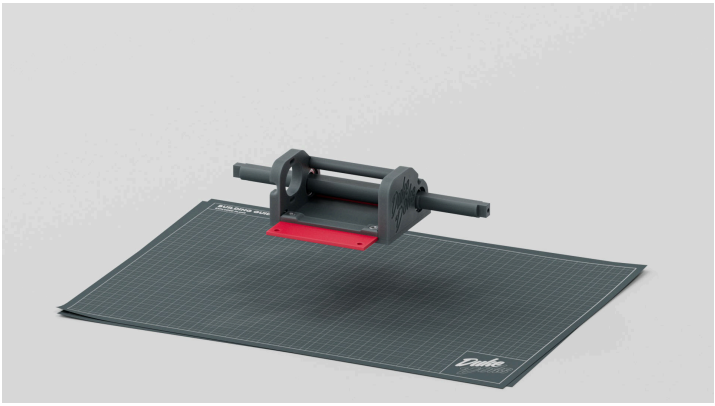
Rear block. We place the bearings in “Rear-Frame_L” and “Rear-Frame_R”. We join one of these two to “Rear-Frame-Lower” with two nuts and two 8mm M3 screws. We insert the “Rear-Axis” through the bearing and join the other piece with another two

nuts and two 8mm M3 screws. Next, we join “Rear-Flex-1” and one of the clamps to “Rear-Frame-Lower” at the front (the highest point) with two nuts and two 8mm M3 screws, creating a sandwich. We do the same with “Rear-Flex-2” and another clamp at the rear. Finally, we reinforce the assembly with the “Rear-Frame-Backing” and two 8mm M3 screws.

The "Rear-Frame_R" part is where the motor is mounted. This part works in conjunction with one of the two pinions. There are two versions of these parts; both are good, so you can use whichever works best for you. The simpler one, and the one I recommend, is version "A." The motor mount is flat, and the pinion is attached by inserting two screws along the motor shaft. These screws are a tight fit, exerting force on the shaft to secure the pinion. In version "B," the motor mount has two protrusions to create space for this version of the pinion. This pinion has two holes for inserting two nuts, through which two 5mm M3 screws will pass. These screws will be tightened against the motor shaft to secure it. In this step, we only need the "Rear-Frame_R" part, but you will need to use either "A" or "B" depending on which pinion you are using: "Gear-1_A" or "Gear-2_B."





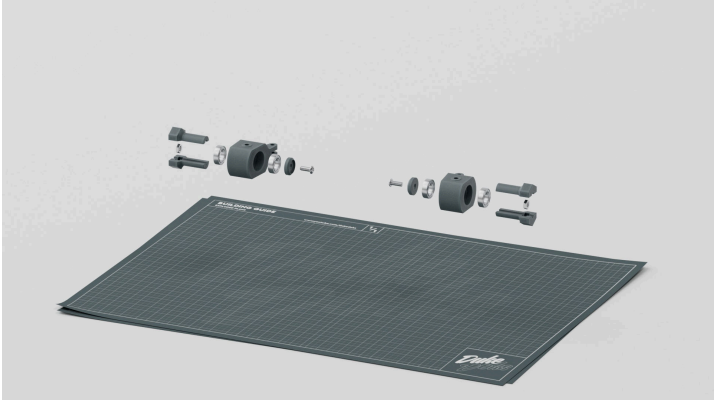


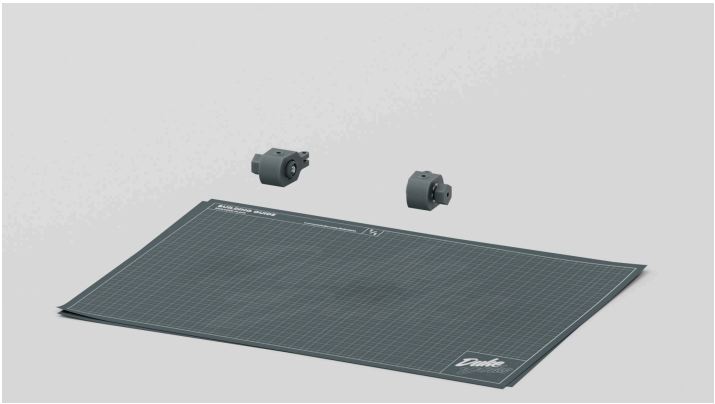
Step 3

Components: 2 x 8mm M3 screws, 2 x M3 nuts and 4 x 8x12x3.5mm bearings

Printed parts: “Wheel-Hub” x2, “Wheel-Axis” x4 and “Wheel-Axis-Nut” x2

Front axles. We fit one bearing on each side into the "Wheel-Hub" pieces. Then, we form each axle by joining two "Wheel-Axis" pieces and placing an M3 nut in the inner recess. We insert the assembly through the bearings and secure it at the other end with the "Wheel-Axis-Nut" and screws.





Step 4A

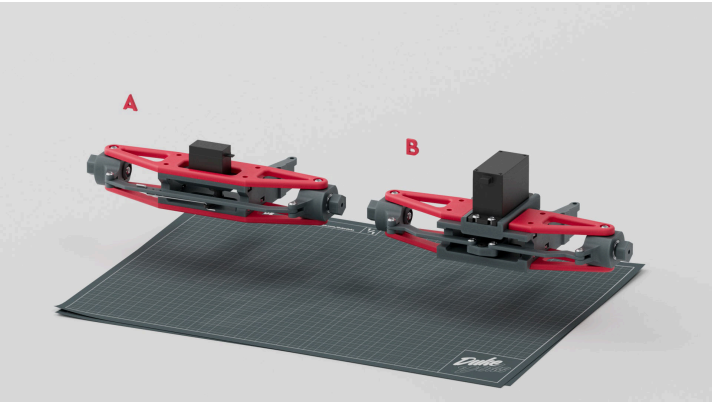
Front Block. In this step, we have two options. Step 4A corresponds to front block "A," which has a small servo for steering. If you want to build the chassis with this front block, stay at this step and skip the next one. However, if you want it with the large servo (which is what I recommend), skip this step and continue to Step 4B. The files specifically for this block have the prefix "A" in their name.

Components: 4 M3 nuts, 6 M3 8mm screws and a piece of wire or a clip.

Printed parts for small servo: “A_Front-Bars-Upper”, “A_Front-Frame” and “A_Steering”

Common printed parts for both front block versions: “Front-Bars-Lower” and Step 3 front axles

Front block “A”. Fit the 8 M3 nuts into the holes in “A_Front-Frame”. Next, attach the 2 front axles to “Front-Bars-Lower” with 2 x 8mm M3 screws and place “A-Front-Frame” on top, but do not screw it in. Then, attach “A_Front-Bars-Upper” by screwing it only to the side axles. Now, connect these axles together with “A_Steering” and 2 x 8mm M3 screws. Finally, place the servo with its screws in the upper hole and attach it to “A_Steering” with a piece of wire using the holes in the servo arm and the steering piece. This piece of wire can be a paperclip. Bend it as shown in the images to keep it in place.





Step 4B

Front Block. If you built the front block with the small servo, skip this step. If you want to build it with the large servo, continue. The files specifically for this block have the prefix "B" in their name.

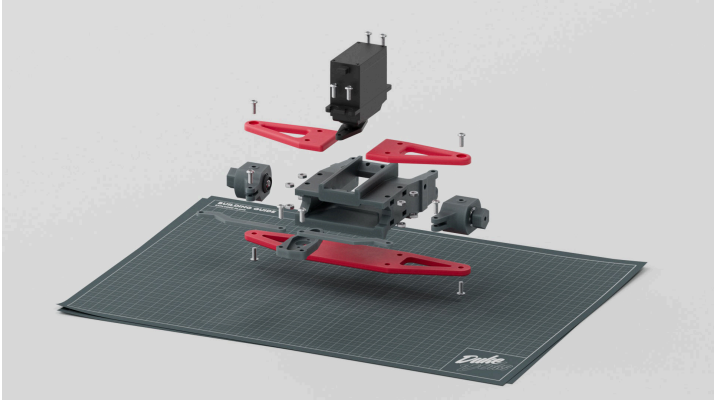
Components: 4 M3 nuts and 8 M3 screws of 8mm

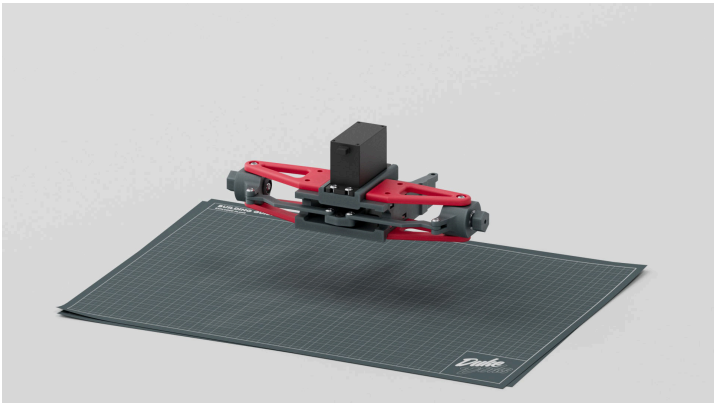
Printed parts for large servo: “B_Front-Bars-Upper L”, “B_Front-Bars-Upper_R”, “B_Front-Frame”, “B_Servo-Arm”, “B_Steering_L” and “B_Steering_R”

Common printed parts for both front block versions: “Front-Bars-Lower” and Step 3 front axles

Front Block B. Fit the eight M3 nuts into the holes in “B_Front-Frame”. Next, attach the two front axles to “Front-Bars-Lower” with two 8mm M3 screws and place “B-Front-Frame” on top, but do not screw it in. Then, attach “B_Front-Bars-Upper_L” and “B_Front-Bars-Upper_R”, screwing them only to the side axles. Separately, prepare the servo arm by attaching “B_Steering_L” and “B_Steering_R” to “B_Servo-Arm” with two 8mm M3 screws. Attach this assembly to the servo arm and fit everything through the upper hole in “B_Front-Frame”. Now we fix the servo in

place with 4 M3 8mm screws and attach the steering arms “B_Steering_L” and “B_Steering_R” to the shafts with another two M3 8mm screws.



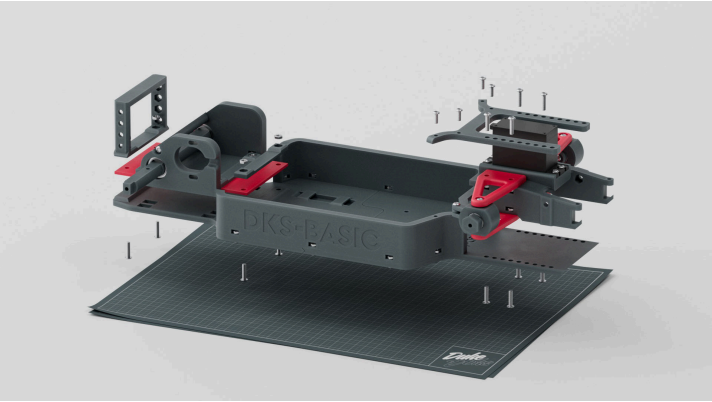


Step 5

Components: 4 M3 screws of 8mm and 12 M3 screws of 12mm.

Printed parts: “Clamp”, “Clamp-Case-Holder”, “Front-Bars”, main body, front block and rear block.

Chassis. We attach the rear block to the main body using four 12mm M3 bolts, four M3 nuts, a clamp, and a clamp case holder, creating two sandwiches. The rear block will be attached to the main body by the two flexible pieces that act as dampers. Next, we attach the front block to the main body with four 12mm M3 bolts from below, using the nuts we installed in the previous step. Now we also attach it from above with the front bars and four 8mm M3 bolts that will connect this piece to the main body, and four 12mm M3 bolts that will screw into the front block, again using the nuts from the previous step. At this point, we can choose the wheelbase. Use the holes on the main body and the front bars that best suit your body. For example, for the Challenger body, we should use the furthest position, and for the DeLorean, the closest position, as shown in the images.



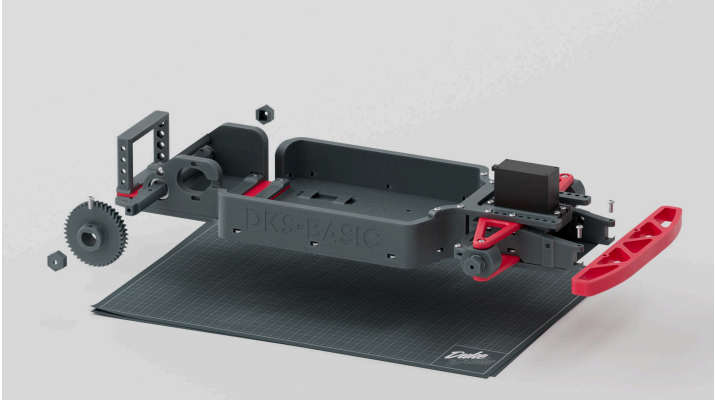


Step 6

Components: 3 x 8mm M3 screws.

Printed parts: “Gear-2”, “Wheel-adapter” x2 and “Bumper”

Bumper and rear axle. We insert "Gear-2" into the rear axle and secure it with an 8mm M3 screw. Then we fit the "Wheel-adapters," one onto each end of the rear axle. Finally, we attach the "Bumper" to the front block with two 8mm M3 screws. This bumper might be inconvenient for some body styles, so it's optional; its only function is to protect the chassis in the event of a frontal collision.





Step 7

Components: 4 x 12mm M3 screws, 4 x M3 nuts and 4 washers.

Printed parts: “Front-Case-Holder” and “Rear-Case-Holder”

Body mounts. We attach the "Front-Case-Holder" to the front block and the "Rear-Case-Holder" to the rear block with two 8mm M3 bolts and two M3 nuts each. It is recommended to use washers to increase the contact surface and prevent the bolt heads from damaging the holes in the parts. These parts will determine the body's height; there are several holes to adjust it to your liking.

Step 8

Components: 2 x 12mm M3 screws, 2 x 16mm M3 screws and 2 nuts.

Printed parts: “Basic-Wheel-01-Wheel” x4, “Basic-Wheel-01-Nut” x4 and “Basic-Wheel-01-Tire” x4.

Wheels. We fit the four "Basic-Wheel-01-Tire" tires onto the four "Basic-Wheel-01-Wheel" rims and attach the four wheels to the chassis axles by placing a "Basic-Wheel-01-Nut" in the middle. The front wheels are attached with two 8mm M3 screws, which fit into the nuts inside the axles. The rear wheels are secured with two 16mm M3 screws and two nuts, which are placed in the holes on the axle. These nuts can be standard or self-locking. This will prevent the wheels from loosening with use.

Step 9

Components: 8 magnets, 8 M3 8mm countersunk screws and 8 M3 nuts.

Magnets. We attach the 8 magnets to the body mounts using the screws and nuts. It's not strictly necessary to use 8 magnets; you can use only 4 and omit the 4 in the center. Using 8 simply increases the strength with which the chassis attaches to the body.

Step 10

Components: Motor, 2 M3 screws of 8mm and 2 M3 screws of 12mm.

Printed parts: “Gear-1”

Motor. Insert gear "Gear-1" onto the motor shaft and screw in the two 8mm M3 screws. This will cause the plastic to expand, making the shaft much tighter. Use a cloth or something similar to hold the gear in place. Now position the assembly and secure it with the two 12mm M3 screws. You can also use two washers here to avoid putting too much stress on the elongated holes of the printed part.

As you saw in step 2, the gear you can use will depend on your decision in that step. If you chose version “B” instead of two 8mm M3 screws, you will need two 5mm M3 screws and two nuts. Go back to step 2 to see the image.

Step 11

Now it's time to attach the rest of the components. I've simply used Velcro straps to secure the battery and adhesive Velcro for easy removal and replacement of the components. If your components are different, your battery is larger, or you can't position them exactly as I did, don't worry, arrange them to your liking. You can use version "A" of the chassis and utilize the holes to create custom brackets and mounts for your components. See step 1.

You can also add a body; I've already designed some, one of a Dodge Challenger, another of a DeLorean, etc. You can see their construction guides on this same website; the one for the Challenger, for example, you can see it here [https://dukedoks.com/portfolio/guia-dodge-challenger/] .

As for connecting the components, it's very simple. They're typical radio-controlled components. You'll find thousands of videos and guides about them online. Here [https://rcfanatic.es/conectar-la-electronica-del-coche-radio-control/] 's a link to a website that explains everything in detail.

Last step

Share your chassis with me. One of the things I love most about designing is seeing others print and enjoy my designs. Share photos and videos of your chassis, the finished product, or some drifts on social media and tag me @dukedoks. I'd love to see this car around the world! And if you've made it this far, thank you so much. I hope you've enjoyed this project as much as I have. See you in the next project!

More Projects





